

Being Sorry is the Hardest Thing: How Robots can Apologize and Learn from Mistakes to Restore People's Trust

Hideki Garcia Goo
Human Media Interaction
University of Twente
Enschede, Netherlands
h.garciagoo@utwente.nl

Bob R. Schadenberg
Human Media Interaction
University of Twente
Enschede, Netherlands
b.r.schadenberg@utwente.nl

Jan Kolkmeier
Human Media Interaction
University of Twente
Enschede, Netherlands
jankolkmeier@gmail.com

Khiet P. Truong
Human Media Interaction
University of Twente
Enschede, Netherlands
k.p.truong@utwente.nl

Vanessa Evers
Human Media Interaction
University of Twente
Enschede, Netherlands
Nanyang Technological University
Singapore, Singapore
vanessa.evers@ntu.edu.sg

Abstract

Robots that move around people in the workplace will make mistakes, such as getting too close to someone or blocking the way. These mistakes negatively impact people's experience with and attitudes toward robots. For a robot to successfully recover from mistakes during navigation, we need deeper insights into the effects of these mistakes and the effectiveness of different recovery strategies. We conducted an online survey (N=219) to assess people's trust in a robot before and after errors, using three recovery strategies (communicating learning capability, apologizing, or a combination of both). Results show that trust dropped significantly after a navigational error but could be effectively restored, especially when the robot apologizes and communicates its ability to learn from it. When the robot does not try to recover from a mistake, the effects are very detrimental. These results highlight the importance of appropriate error recovery strategies for robots to enhance human-robot interactions.

CCS Concepts

• **Human-centered computing** → **Empirical studies in interaction design**; • **Computer systems organization** → *Robotics*.

Keywords

robots, trust recovery, mitigation strategies, robot errors, navigation, healthcare

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1 Introduction

Service robots are increasingly transitioning from controlled industrial or research environments to real-world settings shared with people—such as hospitals, airports, restaurants, and farms. In these social and semi-structured spaces, navigation is essential for smooth interaction with nearby humans [26, 39]. However, robots inevitably make errors when maneuvering around dynamic crowds, animals (i.e., livestock), or other moving machines (i.e., carts, farm machinery). Prior work shows that such mistakes can undermine users' trust and acceptance [1, 16, 20, 36].

To mitigate the negative effects of these errors, researchers have adapted common human trust repair strategies, such as apologies or assurances, to robots [5, 13]. While apologies can promote trust recovery in certain contexts [7, 8], they are not universally effective [12] and may shift responsibility away from developers [22]. Machine-inspired approaches, such as error messages [32] or automatically generated apologies [33], are less studied in HRI but may align better with a robot's unique communication modalities (e.g., sounds, lights) [14]. Yet it remains unclear which strategies—or combination of strategies—work best, and whether effectiveness depends on factors such as error type or severity.

In this study, we investigate how robots can recover trust after making *navigational errors* in a hospital-like context. We compare two distinct recovery strategies of which one is more human-like (apologising) one more machine-like (learn from error), their combination, and no error mitigation at all, examining which approach best attenuates the negative consequences of mistakes. Specifically, we explore whether a social, human-inspired method (e.g., apology) outperforms a robot-specific strategy (e.g., communicating learning capabilities) or whether combining both yields stronger trust repair. We begin by reviewing relevant literature on robot errors and mitigation strategies, followed by our problem statement, aims, and hypotheses. We then describe our online, video-based survey study, present quantitative results, and discuss the implications of

our findings for the design of error recovery behaviors in service robots.

2 Background

Robot errors are inevitable in HRI, and understanding both their nature and effects is crucial for improving interaction design. Numerous studies have observed negative outcomes from mistakes in diverse applications. For instance, serving the wrong food [40] or delivering incorrect orders [24] undermines user satisfaction, trust, and perceptions of robot competence. Although some research suggests that certain errors (e.g., giving faulty LEGO instructions) can increase likability [27], most work finds that mistakes (e.g., erratic navigation [31], dropping objects [30, 34], and social norm violations [3]) generally diminish users' experiences and trust in robots [1, 16, 25] and can discourage their adoption if expectations are not met [36].

Scholars typically distinguish between social norm violations and technical or performance errors [17, 20, 35], though these can overlap from a user's perspective (e.g., a sensor failure leading to personal space intrusion). Tian and Oviatt [35] found that social errors reduce perceived social intelligence, whereas performance errors erode trust in competence. Factors such as the robot's perceived anthropomorphism [8, 13], error perceived severity [34], and the specific violation type [13] can further influence how strongly errors affect people. Given these consistently negative impacts on attitudes and trust [20], it is critical to investigate whether—and how—robots can recover from the detrimental consequences of making mistakes.

When robots make mistakes, it can violate the trust of users [1, 16, 20, 28, 36]. Research on interpersonal trust violations indicates that rebuilding trust is difficult and time-consuming; similarly, in human-robot interaction, trust is hard to earn but easy to lose. As Esterwood [11] notes, "human-robot trust repair strategies should fully restore trustworthiness and its various sub-dimensions: ability, benevolence, and integrity after repeated trust violations."

Many studies on robot errors and mitigation strategies (e.g., [13, 20, 35, 36]) explore trust restoration (e.g., [1, 4, 7, 9, 12, 18, 19, 24, 25, 29, 33]), take inspiration from behaviours in human interactions. Often examining apologies [7, 12, 19, 29, 33], denials [12, 33], explanations [7, 12, 25], and promises [12, 29]. Researchers have also emulated human remorseful behaviour when making a mistake into robot behaviours by adding anthropomorphic expressions [19, 28].

Although error mitigation strategies have been successful in restoring trust, their effectiveness is not universal. In some cases, they fail, or certain strategies prove more effective than others depending on the error type and context. Additionally, individual user characteristics (e.g., gender, personality traits), perceived error severity [4, 41], robot anthropomorphism [10], and timing of the mitigation strategy [13, 21, 29, 30] influence both the preference for and the effectiveness of trust recovery strategies [6, 24]. This variability highlights the importance of employing context-sensitive trust recovery strategies that account for error-specific and user-specific factors.

3 Problem Statement and Hypotheses

This research, part of larger projects on service robots in healthcare and agricultural environments, primarily examines how people respond to robot mistakes and how robots can mitigate their negative effects. In workspaces, robots will inevitably make or appear to make errors, especially during complex interactions like overtaking groups or reacting to bystanders. Since it's impractical to design robot behaviors for all social scenarios, robots are often trained through trial and error (reinforcement learning). They learn from positive and negative reinforcement—such as people's responses or facial expressions—to identify inappropriate and appropriate responses in various situations [3, 37]. This optimization process inevitably leads to mistakes that negatively impact how people work with, experience, and trust the robot. Trust violations may be repaired, but the effectiveness of repair strategies varies: human-inspired approaches (e.g., apologies) might not translate well to human-robot interaction, while machine-inspired strategies (e.g., demonstrating autonomous learning from mistakes) may require users to seek proof that errors won't recur. Given these uncertainties, it is crucial to determine whether trust repair strategies enable robots to effectively recover from mistakes, whether human strategies are effective for robots or if machine-inspired strategies are preferable, and whether different types of errors require different repair methods.

In this study, we investigate how human-inspired social error recovery strategies, robot-inspired technical strategies, and their combinations affect people's trust in robots across various error scenarios in human-robot interactions.

RQ1. To what extent do navigation errors affect people's trust in a service robot?

Prior research shows that robot errors often lower trust [35]. Thus, we propose: **H1a:** Participants' trust in the robot will decrease following a navigation error. Severity also matters: more severe errors typically elicit stronger negative responses [30]. Hence: **H1b:** The perceived severity of a navigation error will negatively influence users' trust, with more severe errors having a more negative impact.

RQ2. To what extent can a service robot recover from navigation errors using different error recovery strategies?

Some studies suggest that social strategies like apologizing are most effective [13], but it remains unclear whether more technical, robot-specific strategies might work better for robots. We hypothesize: **H2a:** The type of error recovery strategy will affect trust repair, the robot learning from mistakes will be most effective to recover from a navigation error. Prior work indicates that the size of the trust violation influences the ability to recover trust [10]. Thus: **H2b:** For larger trust violations the robot's error recovery strategies will be less effective in repairing trust.

RQ3. To what extent does the perceived severity of a robot's error influence the effectiveness of different mitigation strategies for trust repair?

Evidence shows that severity of the error shapes the effectiveness of any mitigation attempt [4, 9, 41], with technical approaches being particularly helpful when severity is high [4]. Therefore: **H3:** The severity of the error will determine the preferred mitigation strategy: For severe errors the 'technical' learning strategy will

work best and for less severe errors the 'social' apologizing strategy will be best to repair trust.

This study used an online survey to explore the influence of error severity and trust repair strategies on people's trust in the robot.

4 Method

4.1 Study Design

We used a 4×3 mixed-design, with four within-subjects Mitigation Strategies (apology, learning, both, none) and three between-subjects Error Conditions (abrupt stops, suddenly coming around the corner, not keeping sufficient distance). Participants were randomly assigned to one Error Condition (74 to abrupt, 73 to corner, 72 to distance) and experienced all four Mitigation Strategies in randomized order.

4.2 Participants

Participants were recruited via Prolific, received €8 for their time, and completed the survey in about 20 minutes ($M = 20.38$, $SD = 12.91$). Only European residents proficient in English and using audio-enabled computers were included. The study was approved by the Ethics Committee (anonymized for review).

Most participants were male (58%), followed by female (40%), non-binary (2%), or prefer not to say (< 1%). The average age was 28.7 years ($SD = 9.28$), with moderate healthcare experience ($M = 5.19$, $SD = 2.02$) and moderate robot familiarity ($M = 5.05$, $SD = 1.54$). Regarding nationality, most were from Portugal (24.7%) or Poland (23.7%), with smaller groups from Spain (8.2%), Italy (6.8%), Greece (5.5%), Hungary (3.2%), Germany (2.7%), Czech Republic (2.7%), Estonia (1.4%), Sweden (1.4%), and other European countries (19.6%). One participant (0.5%) did not disclose nationality.

4.3 Stimuli

We created brief Unity-based animations of a healthcare robot navigating a hospital corridor-like environment and encountering hospital staff (virtual human). Three 8-second videos depicted robot errors (shown in Figure 1): *Abrupt stop*—the robot suddenly halted as a staff member walked closely behind it; *Corner error*—the robot unexpectedly turned toward a staff member approaching a corridor corner; *Distance error*—the robot drove uncomfortably close to a distracted staff member on their cellphone. Each error was paired with one of four trust recovery responses: (1) human-inspired apology (apology; sad eyes, remorseful sound; 23 seconds), (2) machine-inspired message (learning; error message, communication of learning capabilities; 31 seconds), (3) combined human-machine inspired communication (apology and learning; 33 seconds), or (4) no mitigation (14 seconds). Figure 2 shows the combination. Baseline videos (14 seconds) showed the robot navigating correctly. Videos are provided as supplement materials.

4.4 Measures

The **Multi-Dimensional Measure of Trust (MDMT)** is a 16-item scale for human-robot and human-human contexts. It assesses four trust dimensions—Reliability ($\alpha = .92$, Capability ($\alpha = .92$, Ethics ($\alpha = .81$, and Sincerity ($\alpha = .79$ for Sincere)—grouped into two factors:



Figure 1: Screen capture of video animations showcasing the robot error. Left to right: abrupt, corner, and distance errors

capacity trust (Reliable, Capable) and moral trust (Ethical, Sincere). We focus on the capability subscale (capable, skilled, competent, meticulous)[38], but removed “meticulous,” added “intelligence,” and introduced “trustworthy.”

Additionally, to measure the perceived **Error Severity**, we asked participants to rate the perceived severity of the error presented in the video (“In your opinion, how severe was the mistake the robot made? The mistake is...”) with a bipolar 7-point scale from *minor* to *severe*.

4.5 Procedure

Data collection occurred online via Qualtrics and Prolific. After consenting, participants viewed two baseline videos featuring a healthcare delivery robot in a corridor: one showcasing social capabilities (e.g., greeting), and one showing correct robot behavior (no errors). They then answered open-ended questions describing what they saw in the video and completed nine 7-point Likert items (4 Perceived Social Intelligence, 5 Trust), along with a question about whether they think it is a good or bad idea for the hospital to allocate donation funds toward purchasing the robot.

Next, participants viewed an error video and again provided open-ended responses about the video, rated error severity, and repeated the PSI and Trust items plus the donation question. They then saw four error-mitigation videos in a randomized order, each followed by open-ended questions, three Likert statements (robot's

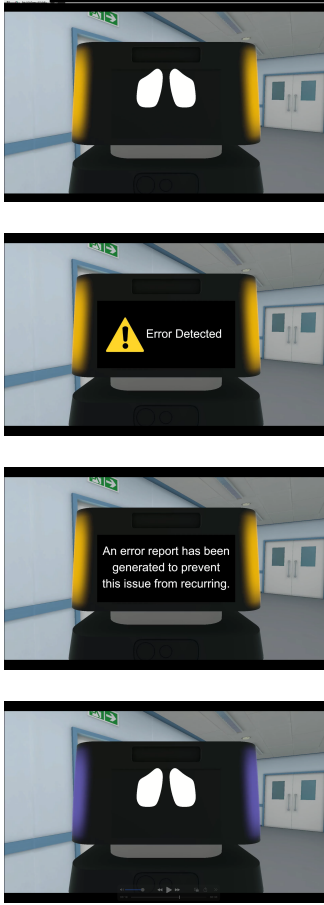


Figure 2: Screen captures of the robot’s combined Apology and Learning mitigation strategy. Left to right: zoom on regretful eye expression, display error message, show learning message, zoom out with regretful eye expression.

remorse, awareness, and future error likelihood), and the same nine Likert items plus the donation question. Finally, demographic data (e.g., age, gender, country, robot familiarity) was collected.

4.6 Analysis

We used mediation in R to test how error severity and error type affect Trust and Perceived Social Intelligence (PSI). A Linear Mixed Model (LMM) via lme4 [2] and lmerTest [23] then accounted for fixed effects (Error, Mitigation, Initial Loss of Trust, Age, Gender, Healthcare Experience, Robot Familiarity) and random effects (participant ID), following [15] for model selection (BIC-based). Trust was the dependent variable (see Section 4.4). We also used mediation to check whether perceived severity mediates the link between error type and our outcomes, and computed Cronbach’s Alpha (psych package) and descriptive statistics for scale reliability.

5 Results

5.1 Reliability of the Trust Scale

Reliability analysis for Trust (skilled, capable, intelligent, competent, trustworthy) revealed high internal consistency ($\alpha = 0.94$), indicating measurement of a single underlying construct.

5.2 Effects of Errors on Trust

A paired t-test showed that trust was significantly lower after the robot made a navigation error than before ($t(218) = -19.07, p < .001$), with an average drop of 1.55 points. A weak correlation ($r=0.35$) between error severity and trust loss suggests more severe errors produce somewhat larger declines in trust.

5.3 Effects of Errors on Trust Mediated by Severity

Descriptive statistics (see Table 3 in Appendix A) provide an overview of the differences in trust values across various error types and mitigation strategies. These statistics highlight how participants’ trust was affected by different types of errors and the mitigation strategies applied (See Figure 3).

5.4 Effect of Mitigation Strategies Across Errors

Descriptive statistics (see Table 4 in Appendix A) illustrate differences in trust values across error types and mitigation strategies (see Figure 3). To determine the most appropriate statistical model, we employed a hierarchical model selection approach based on Bayesian Information Criterion (BIC), following [15].

Table 1 presents fit statistics for sequential model comparisons. We started with a baseline model (Model 1) including only mitigation strategies. Adding error type (Model 2) improved fit ($\chi^2 = 6.96, p = .030$). Model 3 introduced initial loss of trust, significantly improving log-likelihood and reducing AIC and BIC ($\chi^2 = 151.27, p < .001$), confirming its role in trust dynamics. Model 4 tested age, showing a marginal effect ($\chi^2 = 3.90, p = .048$), but BIC increased, indicating uncertainty in its contribution. Given the small improvement, we opted to exclude age, favoring the more stable Model 3.

We then explored interactions between the Mitigation and Error (Model 5), but the fit remained similar to simpler models. However, adding initial trust loss to the interaction model (Model 6) significantly improved fit ($\chi^2 = 151.27, p < .001$). The final model (Model 7) included interactions between mitigation, error type, and initial trust loss, achieving the best fit with the lowest AIC (2183.60) and BIC (2269.50) and the highest log-likelihood improvement ($\chi^2 = 88.67, p < .001$). While Model 3 was already strong, Model 7 better explained how mitigation effects vary across error types and loss of trust. The improvement over simpler models was small but meaningful, justifying its selection while avoiding overfitting.

The results of the linear mixed-effects model (Model 7 in Table 1) analyzing trust as an outcome are presented in Table 2.

Effect of Mitigation Strategy. Compared to the reference level (both apology and learning), no mitigation significantly decreased trust ($\beta = -0.81, SE = 0.14, t(645) = -5.78, p < .001$). Learning also showed a significant negative effect ($\beta = -0.30, SE = 0.14, t(645) =$

Table 1: ANOVA sequential comparisons for Trust model building.

<i>Model</i>	<i>df</i>	<i>AIC</i>	<i>BIC</i>	<i>Log-Likelihood</i>	χ^2	<i>p</i>
Basic models						
(1) Mitigation Only	6	2422.30	2451.00	-1205.20	-	-
(2) Mitigation + Error	8	2419.30	2457.60	-1201.70	6.96	.030*
(3) Mitigation + Error + Initial Loss of Trust	9	2270.10	2313.10	-1126.00	151.27	<.001***
(4) Mitigation + Error + Initial Loss of Trust + Age	10	2268.20	2315.90	-1124.10	3.90	.048*
Interaction models						
(5) Mitigation $\times$ Error	14	2415.50	2482.40	-1193.80	-	-
(6) Mitigation $\times$ Error + Initial Loss of Trust	15	2266.20	2337.90	-1118.10	151.27	<.001***
(7) Mit. $\times$ Error + Initial Loss of Trust $\times$ Mit.	18	2183.60	2269.50	-1073.80	88.67	<.001***

$-2.16, p = .032$), while apology did not have a significant effect ($\beta = -0.26, SE = 0.14, t(645) = -1.87, p = .061$).

Effect of Error Condition. Compared to the reference level (abrupt error), neither corner errors nor distance errors significantly affected trust.

Effect of Initial Loss of Trust. The main effect of initial loss of trust was highly significant, showing that greater initial loss of trust predicted increased trust recovery ($\beta = 0.77, SE = 0.05, t = 14.58, p < .001$).

Interaction Effect of Mitigation Strategy and Error Condition. The interaction effects between Mitigation and Error Condition were non-significant.

Interaction Effect of Mitigation Strategy and Initial Loss of Trust. The interaction between apology and initial loss of trust was significant ($\beta = -0.15, SE = 0.06, t(645) = -2.68, p = .008$), indicating reduced trust recovery for participants with a larger initial loss when the robot apologized. Similarly, no mitigation showed a highly significant negative interaction ($\beta = -0.51, SE = 0.06, t(645) = -8.97, p < .001$), reflecting a strong drop in trust recovery without mitigation.

Using Tukey's post-hoc comparisons, trust was highest with both apology and learning ($b = 1.74, SE = 0.07, p < .001$), followed by learning alone ($b = 1.445, SE = 0.073, p < .001$) and apology alone ($b = 1.27, SE = 0.073, p < .001$). The combination also significantly surpassed apology alone ($p < .001$).

By error type, both apology and learning significantly increased trust compared to no mitigation for abrupt ($b = 1.49, SE = 0.12, p < .001$), corner ($b = 1.74, SE = 0.13, p < .001$), and distance ($b = 2.00, SE = 0.13, p < .001$) errors. Differences between apology and learning alone were generally non-significant (p values = .082–.973), suggesting that combining human (apology) and machine (learning) strategies consistently provided the strongest trust recovery (see Figure 3).

6 Discussion and Limitations

Our findings strongly support **H1a**, demonstrating that navigation mistakes significantly reduce trust in a robot [21, 24, 30, 31]. Although **H1b** predicted a more pronounced trust drop for severe errors, we observed only a modest link between perceived severity and trust loss [30]. This discrepancy may stem from the study's online video format, which could dampen emotional engagement and thus lessen the perceived impact of error severity. Regarding

H2a, the robot's type of mitigation strategy significantly influenced trust recovery. Communicating both apology and learning was the most effective approach, followed by learning alone and then apology alone. This suggests that integrating social (apology) and technical (learning) strategies provides a more comprehensive trust repair mechanism, addressing both emotional and rational aspects of trust. **H2b** posited that larger trust violations would be harder to repair [10]; however, our results indicate that even severe violations can be effectively mitigated if the robot demonstrates both remorse and a plan to learn. This highlights the potential of combined mitigation strategies to overcome substantial trust deficits. **H3** predicted that error severity would shape the preferred mitigation strategy [4, 9, 41]. Consistent with this hypothesis, the "technical" learning strategy became more critical as error severity increased. Nonetheless, a combination of apologising and learning remained the most effective approach across all error types, underscoring the value of a dual-strategy in trust restoration.

While our study focused on service robots in a hospital corridor-like environment, the findings extend to any environment where robots must navigate around humans—such as manufacturing floors or greenhouses. For instance, in farming settings, a robot that inadvertently collides with equipment or comes too close to a worker may benefit from recovery strategies that communicate both social awareness (e.g., an "apology" to acknowledge the violation) and machine-specific assurances (e.g., notifying that it has "learned" to adjust its trajectory). These integrated strategies can help restore trust by addressing both the human and technical dimensions of the error.

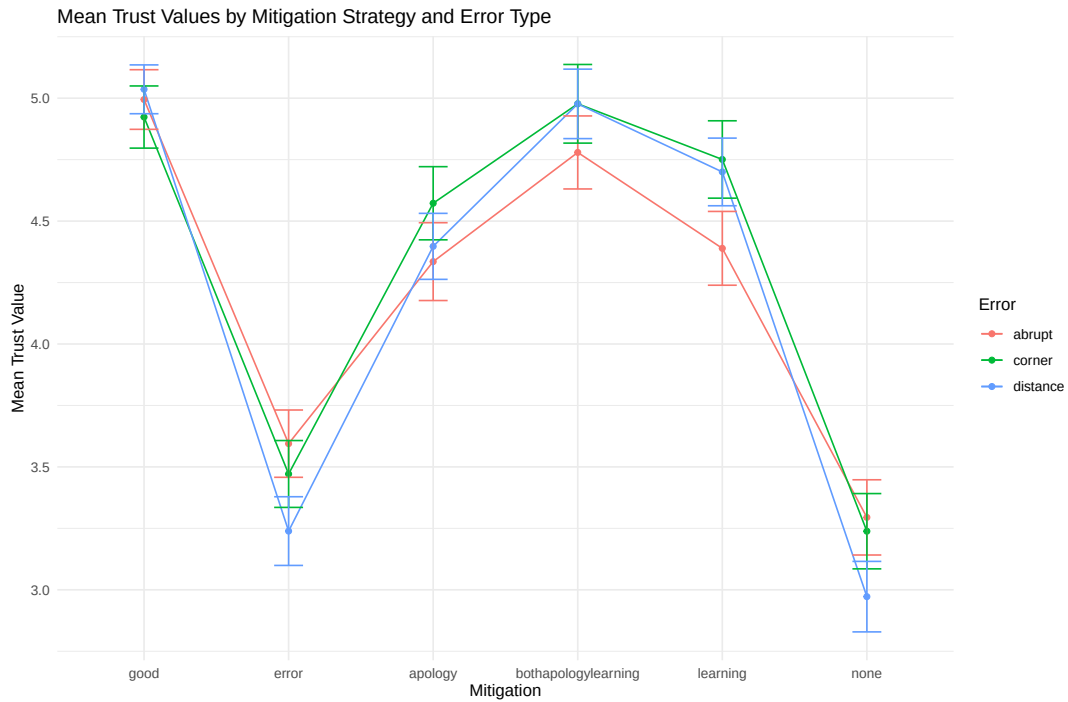
Despite these insights, our study has limitations. Most notably, the video-based format limits ecological validity. Participants passively observed robot errors rather than experiencing them first-hand, which could have influenced their emotional reactions, perceived severity, and trust dynamics. The findings are based on a European sample and specific videos, errors, robots, and mitigation strategies, which may limit generalizability.

Future research should assess generalizability across different cultures, real-world settings and how well trust recovery strategies were conveyed, and a comparative study between verbal and non-verbal communication of these strategies. Exploring trust repair in environments like agriculture—where robots navigate uneven terrain, interact with livestock, or work amid harvest crews—can determine if combined social and technical recovery strategies remain effective across diverse contexts and error types. Studies should

Table 2: Trust: Mitigation Strategy, Error Condition, and Their Interaction.

<i>Effect</i>	<i>Estimate (b)</i>	<i>SE</i>	<i>t-value</i>	<i>p-value</i>
Intercept	0.14	0.13	1.04	.297
Mitigation: Apology	-0.26	0.14	-1.87	.061
Mitigation: Learning	-0.30	0.14	-2.16	.032*
Mitigation: None	-0.81	0.14	-5.78	<.001***
Error: Corner	0.27	0.15	1.78	.075
Error: Distance	0.24	0.15	1.55	.123
Initial Loss of Trust	0.77	0.05	14.58	<.001***
Mitigation: Apology x Error: Corner	0.06	0.16	0.34	.731
Mitigation: Learning x Error: Corner	0.18	0.16	1.08	.279
Mitigation: None x Error: Corner	-0.22	0.16	-1.34	.180
Mitigation: Apology x Error: Distance	-0.07	0.17	-0.41	.685
Mitigation: Learning x Error: Distance	0.16	0.17	0.94	.348
Mitigation: None x Error: Distance	-0.31	0.17	-1.88	.060
Mitigation: Apology x Initial Loss of Trust	-0.15	0.06	-2.68	.008**
Mitigation: Learning x Initial Loss of Trust	-0.08	0.06	-1.50	.134
Mitigation: None x Initial Loss of Trust	-0.51	0.06	-8.97	<.001***

Note: $R^2_{\text{marg.}}$: .740; $R^2_{\text{cond.}}$: 0.544

**Figure 3: Model Trust per Error before ('good'), after ('error') trust violation, and across the four different mitigation strategies**

also examine how initial trust loss impacts mitigation strategies and include repeated interactions to observe trust evolution over time with multiple mistakes. Incorporating objective behavioral measures (e.g., reliance on or avoidance of the robot) will provide deeper insights into trust development and repair. We plan to analyze additional variables, such as willingness to donate, perception of the robot's social intelligence, and qualitative feedback on robot

behavior, to gain a comprehensive understanding of trust repair processes.

Overall, our results suggest that combining human-inspired and machine-inspired mitigation strategies effectively restores user trust after robot errors. This highlights the importance of designing multifaceted error recovery behaviors in service robots to enhance

human-robot interactions and promote user acceptance across diverse domains.

7 Conclusion

This study advances our understanding of how robots can recover from errors in human-robot interactions, particularly in work settings. Robot mistakes typically reduce trust, yet combining human-inspired (apology; non-verbal show of regret) and machine-inspired (learning; error message and show of learning capabilities) trust recovery strategies effectively restores user confidence. Apologies add a social element, while learning capabilities help regain trust. Limitations include a focus on short-term error recovery and the use of virtual simulations rather than real-world tests. Future research should explore long-term trust dynamics and apply these strategies across different user demographics. The major contribution lies in demonstrating the value of blending human and machine approaches to error recovery, thus expanding the HRI design space.

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A Descriptive statistics

A.1 Severity values for errors

Table 3 summarizes the severity statistics for different error types.

A.2 Trust

Table 4 summarizes trust ratings across different mitigation strategies and error types.

Table 3: Summary Severity statistics of Mitigation and Error types.

Mitigation	Error	Count	Mean	Median	SD	Min	Max
Error	Abrupt	74	3.76	3.00	1.70	1	7
Error	Corner	73	4.37	5.00	1.65	1	7
Error	Distance	72	5.36	6.00	1.45	2	7

Table 4: Summary statistics ordered by Good, Error, Apology, Apology and Learning, and Learning on Trust.

Mitigation	Error	Count	Mean	Median	SD	Min	Max
Good	Abrupt	74	4.99	5.20	1.04	1.00	6.80
Good	Corner	73	4.92	5.00	1.08	1.20	6.80
Good	Distance	72	5.04	5.00	0.84	2.80	7.00
Error	Abrupt	74	3.59	3.60	1.18	1.00	6.20
Error	Corner	73	3.47	3.60	1.16	1.00	6.00
Error	Distance	72	3.24	3.20	1.19	1.00	6.00
Apology	Abrupt	74	4.34	4.60	1.36	1.00	6.60
Apology	Corner	73	4.57	4.80	1.27	1.00	7.00
Apology	Distance	72	4.40	4.60	1.14	2.00	6.20
Both Apology Learning	Abrupt	74	4.81	5.00	1.25	1.00	7.00
Both Apology Learning	Corner	73	5.00	5.40	1.34	1.00	7.00
Both Apology Learning	Distance	72	5.00	5.00	1.15	2.00	7.00
Learning	Abrupt	74	4.39	4.80	1.29	1.00	6.40
Learning	Corner	73	4.75	5.00	1.34	1.00	7.00
Learning	Distance	72	4.70	5.00	1.17	1.80	7.00
None	Abrupt	74	3.29	3.40	1.32	1.00	6.20
None	Corner	73	3.24	3.20	1.31	1.00	5.80
None	Distance	72	2.97	3.00	1.22	1.00	6.00